

DIRECTED ENERGY WEAPONS: Less-Lethal Laser Weapons



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Under directed energy weapons, so far we have covered particle beam weapons and high-power microwaves. Let us now take a look at some less-lethal laser weapons.

Directed energy laser systems are generally categorised as non-lethal or, more appropriately, less-lethal directed energy laser weapons and lethal high-energy laser weapons. These include laser dazzlers used in anti-personnel applications for anti-terrorist and counter-insurgency operations and lethal high-energy laser weapon systems aimed at inflicting structural damage to the intended targets at tactical and strategic ranges. High-energy lasers used in anti-sensor applications as a part of electro-optic countermeasure equipment also come under less-lethal directed energy laser systems.

Another important application of high-energy directed energy laser systems, when operated at relatively-lower power levels, is in safe neutralisation of unexploded ordnances with minimised collateral damage. These systems, generally configured around kilowatt-class bulk solid-state lasers or high-power fibre lasers, have amply demonstrated their efficacy in the disposal of unexploded ordnances including improvised explosive devices from a safe distance.

Less-lethal laser dazzlers

A low-intensity conflict (LIC) is the most common form of warfare today, and is likely to be so in the foreseeable future. Data suggests that more than 75 per cent of the armed conflicts since World War II have been of the low-intensity variety. An LIC operation is a military term used for deployment and use of troops and/or assets in situations other than conventional war.

As compared to a conventional war,

in the case of LIC operations, armed forces engaged in the conflict operate at a greatly-reduced tempo, generally with fewer soldiers, reduced range of tactical equipment and limited scope to operate in military manner. Also, use of artillery is avoided in the case of conflicts in urban territories, and use of air power is often restricted to surveillance and transportation of personnel and equipment.

LICs, which pose an alarming threat to national security, is an area of concern for the whole of international community, today. Its scope extends from emergency preparedness and response to domestic intelligence activities to riot and mob control, from combating illegal drug trafficking to protection of critical infrastructure, from handling counter-insurgency and anti-terrorist operations to detection of concealed weapons, from detection and identification of chemical and biological warfare agents to detection of explosive materials.

Laser and optoelectronics technologies play an important role in handling LIC situations. Key advantages of the use of laser technology in such applications are near-zero collateral damage, speed-of-light delivery and potential for building non-lethal weapons. Some of the well-established laser devices in LIC applications include laser dazzlers for close-combat operations, mob/riot control and protection of critical infrastructures from aerial threats.

Operational parameters

A laser dazzler emits a high-intensity laser beam in the visible band, usually in the blue-green region, to temporarily impair the vision of the adversary without causing any permanent or lasting injury or adverse effect to the subject's